

Topological Spaces

1 Topological spaces

Denote $\mathcal{P}(X)$ as the power set of a set X .

Definition 1.1 – Topology and topological space

Let X be a set. A topology on X is a set $\mathcal{T} \subseteq \mathcal{P}(X)$ such that

- (i) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$;
- (ii) if $(U_\alpha)_{\alpha \in I}$ is a collection of elements of $\mathcal{T}$, then $\bigcup_{\alpha \in I} U_\alpha \in \mathcal{T}$;
- (iii) if $(U_\alpha)_{\alpha \in I}$ is a finite collection of elements of $\mathcal{T}$, then $\bigcap_{\alpha \in I} U_\alpha \in \mathcal{T}$.

A set X with a topology $\mathcal{T}$ is called a topological space.

Elements of X are called points. Elements of $\mathcal{T}$ are called open sets. If $x \in U \in \mathcal{T}$, then we say that U is an open neighbourhood of x . Strictly speaking, we should always write $(X, \mathcal{T})$ for a topological space. However, when $\mathcal{T}$ is clear, we can just write X for the topological space and $\mathcal{T}_X$ for its topology.

Definition 1.2 – Continuous function

If $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ are topological spaces, then a function $f : X \rightarrow Y$ is called continuous if for any $U \in \mathcal{T}_Y$, we have

$$f^{-1}(U) = \{x \in X : f(x) \in U\} \in \mathcal{T}_X.$$

Definition 1.3 – Homeomorphism

A function $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ is a homeomorphism if it is continuous and has a continuous inverse. That is, f is a bijection, f is continuous, and f^{-1} is also continuous.

If $\mathcal{T} \subseteq \mathcal{T}'$ are both topologies on X , then we say that $\mathcal{T}$ is coarser than $\mathcal{T}'$, and $\mathcal{T}'$ is finer than $\mathcal{T}$. Equivalently, the function $\text{id} : (X, \mathcal{T}') \rightarrow (X, \mathcal{T})$ is continuous.

If (X, d) is a metric space, recall that $U \subseteq X$ is called open if for any $x \in U$, there exists $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq U$.

Proposition 1.1

If $\mathcal{T}_d = \{\text{the subsets of } X \text{ which are open in this sense}\}$, then $(X, \mathcal{T}_d)$ is a topological space. We call this the topology on X induced by the metric d .

Proof. $\emptyset \in \mathcal{T}_d$ tautologically and $X \in \mathcal{T}_d$ as well. Let $\{U_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}_d$. Let $x \in \bigcup_{\alpha \in I} U_\alpha$. We have $x \in U_\beta$ for some β . As U_β is open, there exists $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq U_\beta \subseteq \bigcup_{\alpha \in I} U_\alpha$, so $\bigcup_{\alpha \in I} U_\alpha$ is open.

Now suppose that I is finite and $x \in \bigcap_{\alpha \in I} U_\alpha$. For each α , there exists $\varepsilon_\alpha > 0$ such that $B_{\varepsilon_\alpha}(x) \subseteq U_\alpha$. Set $\varepsilon = \inf_{\alpha \in I} \varepsilon_\alpha = \min_{\alpha \in I} \varepsilon_\alpha > 0$. So, $B_\varepsilon(x) \subseteq B_{\varepsilon_\alpha}(x) \subseteq U_\alpha$ for all α and hence $B_\varepsilon(x) \subseteq \bigcap_{\alpha \in I} U_\alpha$. $\square$

Example 1.1

The Euclidean metric on $\mathbb{R}^d$ defines a topology, the Euclidean topology. For any $X \subseteq \mathbb{R}^d$, the induced metric on X defines a topology on X too. Sets like $\mathbb{R}^d$, $[0, 1]$ or $(0, 1)$ will usually be considered with this topology.

Proposition 1.2

Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \rightarrow Y$ be a function. f is continuous in the metric space sense if and only if f is continuous in the topological space sense.

Proof. Let $f : X \rightarrow Y$ be continuous in the metric space sense. Let $U \in \mathcal{T}_Y$ be open in Y . Let $x \in f^{-1}(U)$, that is $f(x) \in U$. Let $\varepsilon > 0$ be such that $B_\varepsilon(f(x)) \subseteq U$. By metric continuity, there exists $\delta > 0$ such that if $d_X(x, x') < \delta$, then $d_Y(f(x), f(x')) < \varepsilon$, that is $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$. Hence, $B_\delta(x) \subseteq f^{-1}(B_\varepsilon(f(x))) \subseteq f^{-1}(U)$ so $f^{-1}(U)$ is open and f is continuous in the topological space sense.

Now suppose that $f : X \rightarrow Y$ is continuous in the topological space sense. Fix $x \in X$ and $\varepsilon > 0$. Consider $B_\varepsilon(f(x))$ which is open in Y . Thus, $f^{-1}(B_\varepsilon(f(x)))$ is in $\mathcal{T}_X$. So, there exists $\delta > 0$ such that $x \in B_\delta(x) \subseteq f^{-1}(B_\varepsilon(f(x)))$, that is $d_X(x, x') < \delta$ implies $d_Y(f(x), f(x')) < \varepsilon$. Hence, f is continuous in the metric space sense. $\square$

Definition 1.4 – Convergence

Let $(X, \mathcal{T})$ be a topological space and $(x_i) \subseteq X$ be a sequence. We say that this converges to $x_\infty \in X$ if for all open neighbourhoods $U \ni x_\infty$, we have $x_n \in U$ for all $n \geq N$. That is, only finitely many x_i do not lie in U .

Proposition 1.3

If (X, d) is a metric space with topology $\mathcal{T}_d$, then a sequence (x_i) converges to x_∞ in the metric sense if and only if it converges to x_∞ in the topological sense.

Proof. Suppose it converges in the metric sense to x_∞ , then for all $\varepsilon > 0$, we have $x_n \in B_\varepsilon(x_\infty)$ for all $n \geq N_\varepsilon$. If $U \ni x_\infty$ is an open neighbourhood, it contains $B_\varepsilon(x_\infty)$ for some $\varepsilon > 0$. So, $x_n \in B_\varepsilon(x_\infty) \subseteq U$ for all $n \geq N_\varepsilon$.

Conversely, if it converges in the topological sense to x_∞ , let $\varepsilon > 0$. $x_\infty \in B_\varepsilon(x_\infty)$ so there is an $N = N_\varepsilon$ such that $n \geq N_\varepsilon$ implies $x_n \in B_\varepsilon(x_\infty)$. Hence, it converges in the metric sense. $\square$

Example 1.2

Consider $\mathbb{R}$ and $(0, 1)$ with their Euclidean metrics and topologies.

$$\begin{aligned} (0, 1) &\rightarrow \mathbb{R} \\ t &\mapsto \tan(\pi(t - 1/2)) \end{aligned}$$

is a continuous bijection and its inverse is also continuous so it is a homeomorphism and $\mathbb{R} \cong (0, 1)$. However, they are not isometric. $(\mathbb{R}, d)$ is complete but $((0, 1), d)$ is not.

Example 1.3 – Discrete topology

Let X be a set. The discrete topology is $\mathcal{T}_{\text{discrete}} = \mathcal{P}(X)$. Any function from $(X, \mathcal{T}_{\text{discrete}})$ to any space is continuous. It comes from the discrete metric

$$d(x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y. \end{cases}$$

Notice that $B_{1/2}(x) = \{x\}$ is open so all sets are open.

Example 1.4 – Indiscrete topology

Let X be a set. The indiscrete topology is $\mathcal{T}_{\text{indiscrete}} = \{\emptyset, X\}$. Any function from any space to $(X, \mathcal{T}_{\text{indiscrete}})$ is continuous. This does not come from a metric unless $X = \{x\}$. If $x \neq y$, then $d(x, y) = \varepsilon > 0$ and $B_\varepsilon(x) \not\ni y$. Since $B_\varepsilon(x)$ is a nonempty set, it has to be X which is a contradiction.

Let (x_n) be a sequence in $(X, \mathcal{T}_{\text{indiscrete}})$ and $x_\infty \in X$. $x_n \rightarrow x_\infty$ for any choice of $x_\infty \in X$ since X is the only open neighbourhood of x_∞ and $x_n \in X$.

Example 1.5

Let $X = \{o, c\}$ and $\mathcal{T}_{\text{Sier}} = \{\emptyset, \{o, c\}, \{o\}\}$. This is the Sierpinski space. A continuous function $f : Y \rightarrow (X, \mathcal{T}_{\text{Sier}})$ is exactly an open subset of Y .

Example 1.6 – Cofinite topology

Let X be a set. The cofinite topology is $\mathcal{T}_{\text{cofinite}} = \{X \setminus F \mid F \text{ is a finite subset}\} \cup \{\emptyset\}$.

Example 1.7 – Right order topology

Let $X = \mathbb{R}$. The right order topology is $\mathcal{T}_{\text{ord}} = \{(a, \infty) \mid -\infty \leq a \leq \infty\}$. We can check that this is a topology. Let $\{(a_\alpha, \infty)\}_{\alpha \in I}$ be a collection of elements of $\mathcal{T}_{\text{ord}}$.

$$\bigcup_{\alpha \in I} (a_\alpha, \infty) = \left(\inf_{\alpha \in I} a_\alpha, \infty \right) \in \mathcal{T}_{\text{ord}}$$

and if I is finite, then

$$\bigcap_{\alpha \in I} (a_\alpha, \infty) = \left(\max_{\alpha \in I} a_\alpha, \infty \right) \in \mathcal{T}_{\text{ord}}.$$

Definition 1.5 – Bases and subbases

Let $\mathcal{T}$ be a topology on X .

- (i) A basis $\mathcal{B} \subseteq \mathcal{T}$ for $\mathcal{T}$ is a subcollection such that every element of $\mathcal{T}$ is a union of elements in $\mathcal{B}$.
- (ii) A subbasis $\mathcal{S} \subseteq \mathcal{T}$ is a subcollection such that every element of $\mathcal{T}$ is a union of sets which are finite intersections of elements of $\mathcal{S}$.

Lemma 1.1

Let $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ be a function between topological spaces and $\mathcal{S} \subseteq \mathcal{T}_Y$ be a subbasis. If $f^{-1}(U)$ is open for all $U \in \mathcal{S}$, then f is continuous.

Proof. If $V \in \mathcal{T}_Y$, then $V = \bigcup_{\alpha \in I} V_\alpha$ where $V_\alpha = \bigcap_{\beta \in J_\alpha} U_{\alpha,\beta}$ where J_α is finite and $U_{\alpha,\beta} \in \mathcal{S}$. Then,

$$f^{-1}(V) = \bigcup_{\alpha \in I} f^{-1}(V_\alpha) = \bigcup_{\alpha \in I} \bigcap_{\beta \in J_\alpha} f^{-1}(U_{\alpha,\beta}) \in \mathcal{T}_X$$

by the axioms for a topology. □

Example 1.8

Consider the Euclidean topology on $\mathbb{R}^n$. The collection $\mathcal{B} = \{B_r(x) \mid x \in \mathbb{R}^n, r > 0\}$ is a basis. The collection $\mathcal{C} = \{(a_1, b_1) \times \cdots \times (a_n, b_n) \mid a_i < b_i\}$ is also a basis. The countable collection $\mathcal{QB} = \{B_q(x) \mid x \in \mathbb{Q}^n, q \in \mathbb{Q}_{>0}\}$ is also a basis. The Euclidean topology on $\mathbb{R}^n$ is said to be second countable since it has a countable basis.

Definition 1.6 – Closed subset

Let $(X, \mathcal{T})$ be a topological space. A subset $C \subseteq X$ is closed if $X \setminus C \in \mathcal{T}$.

Proposition 1.4

Let $(X, \mathcal{T})$ be a topological space and $\mathcal{F} = \{C \subseteq X \mid C \text{ is closed}\}$. Then,

- (i) $\emptyset \in \mathcal{F}$ and $X \in \mathcal{F}$;
- (ii) if $\{C_\alpha\}_{\alpha \in I}$ is a collection of sets in $\mathcal{F}$, then $\bigcap_{\alpha \in I} C_\alpha \in \mathcal{F}$;
- (iii) if $\{C_\alpha\}_{\alpha \in I}$ is a finite collection of sets in $\mathcal{F}$, then $\bigcup_{\alpha \in I} C_\alpha \in \mathcal{F}$.

Proposition 1.5

A function $f : X \rightarrow Y$ between topological spaces is continuous if and only if the preimage of any closed set is closed.

Definition 1.7 – Closure and interior

Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$ be a subset.

- (i) The closure $\bar{A}$ of A is the smallest closed set containing A . That is, $\bar{A} = \bigcap_{\text{closed sets } C \text{ containing } A} C$. We say A is dense in X if $\bar{A} = X$.
- (ii) The interior $\mathring{A}$ of A is the largest open set contained in A . That is, $\mathring{A} = \bigcup_{\text{open sets } U \text{ contained in } A} U$.

Definition 1.8 – Limit point

Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$ be a subset. A limit point of A is a point in X which is a limit of a sequence in A .

Proposition 1.6

If C is a closed subset of $(X, \mathcal{T})$, then limit points of C lie in C .

Proof. Let (x_n) be a sequence in C with limit x_∞ . If $x_\infty \notin C$, then $x_\infty \in X \setminus C$ which is an open set. If $x_n \rightarrow x_\infty$, then we should have $x_n \in X \setminus C$ for all $n \geq N$. But $x_n \in C$ so $x_n \notin X \setminus C$ and contradiction. □

Corollary 1.1

A limit point of A lies in $\overline{A}$.

Example 1.9

$\overline{\mathbb{Q}} = \mathbb{R}$ since any real number is a limit of a sequence of rationals.

Example 1.10

$\overline{(0,1)} = [0,1]$. Certainly, $\overline{(0,1)} \subseteq [0,1]$. But also $0 \in \overline{(0,1)}$ as it is the limit of the sequence $(1/n)$. Similarly, $1 \in \overline{(0,1)}$ as it is the limit of the sequence $(1 - 1/n)$.

Example 1.11 – Cocountable topology

The cocountable topology on $\mathbb{R}$ is that with $\mathcal{T}_{\text{cocountable}} = \{\mathbb{R} \setminus C \mid C \text{ is countable}\} \cup \{\emptyset\}$. Let (x_n) be a sequence. For any $x \in \mathbb{R}$, consider $U = \{x\} \cup (\mathbb{R} \setminus \{x_n\})$. U is open and contains x . If $x_n \rightarrow x$, then x_n must be in U for all $n \geq N$, so $x_n = x$ for all $n \geq N$. Hence, the convergent sequences are exactly the eventually constant sequences, with their limits being the eventually constant value. So, the limit points of a set A are A . But $(0,1)$ is not closed and $\overline{(0,1)} = \mathbb{R}$ as it is not countable.

Definition 1.9 – Hausdorff

A space $(X, \mathcal{T})$ is Hausdorff if for all $x \neq y \in X$, there are open neighbourhoods $U \ni x$ and $V \ni y$ with $U \cap V = \emptyset$. That is, points can be separated by open sets.

Lemma 1.2

If the topology $\mathcal{T}$ comes from a metric d , then it is Hausdorff.

Proof. If $x \neq y$, then $d(x, y) = \delta > 0$. Consider $U = B_{\delta/2}(x)$ and $V = B_{\delta/2}(y)$. The triangle inequality shows that $U \cap V = \emptyset$. $\square$

Proposition 1.7

If $(X, \mathcal{T})$ is Hausdorff, then sequences in X have at most one limit.

Proof. Let (x_n) be a sequence and suppose $y \neq z \in X$ are both limits. Let $U \ni y$ and $V \ni z$ be disjoint open neighbourhoods. $x_n \in U$ for all $n \geq N_1$ since $x_n \rightarrow y$. Also, $x_n \in V$ for all $n \geq N_2$ since $x_n \rightarrow z$. So, $x_{\max\{N_1, N_2\}} \in U \cap V = \emptyset$ and hence contradiction. $\square$

Proposition 1.8

In a Hausdorff space, points are closed.

Proof. Let $x \in X$, we want to show that $\{x\}$ is closed, that is $\{x\} = \overline{\{x\}}$. Let $y \neq x$. By Hausdorffness, we have $U \ni x$ and $V \ni y$ open with $U \cap V = \emptyset$. So, $x \in X \setminus V$ closed and $\{x\} \subseteq X \setminus V$. But $y \in V$ so $y \notin \{x\}$. Hence, $\overline{\{x\}} = \{x\}$. $\square$

Example 1.12

Let X be an infinite set and consider $(X, \mathcal{T}_{\text{cofinite}})$. Any two non-empty open sets intersect. $(X \setminus F) \cap (X \setminus F') = X \setminus (F \cup F')$ is non-empty since X is infinite but $F \cup F'$ is finite. Hence, it is not Hausdorff.

Definition 1.10 – Subspace topology

Let $(X, \mathcal{T})$ be a topological space and $Y \subseteq X$ a subset. The subspace topology on Y is

$$\mathcal{T}|_Y = \{Y \cap U \mid U \in \mathcal{T}\}.$$

A subspace of $(X, \mathcal{T})$ is a subset equipped with the subspace topology.

Proposition 1.9

The subspace topology is a topology.

Proof. $\emptyset = \emptyset \cap Y \in \mathcal{T}|_Y$ and $Y = X \cap Y \in \mathcal{T}|_Y$. If $\{U_\alpha \cap Y\}_{\alpha \in I}$ is a collection of elements of $\mathcal{T}|_Y$, then $\bigcup_{\alpha \in I} (U_\alpha \cap Y) = (\bigcup_{\alpha \in I} U_\alpha) \cap Y \in \mathcal{T}|_Y$. If I is finite, then $\bigcap_{\alpha \in I} (U_\alpha \cap Y) = (\bigcap_{\alpha \in I} U_\alpha) \cap Y \in \mathcal{T}|_Y$. $\square$

Proposition 1.10

The function $\text{inc} : (Y, \mathcal{T}|_Y) \rightarrow (X, \mathcal{T})$ is continuous. In fact, $\mathcal{T}|_Y$ is the coarsest topology on Y such that inc is continuous.

Proof. Let $U \in \mathcal{T}$, then $\text{inc}^{-1}(U) = U \cap Y \in \mathcal{T}|_Y$ by definition so inc is continuous. Suppose $\text{inc} : (Y, \mathcal{T}') \rightarrow (X, \mathcal{T})$ is continuous. For $U \in \mathcal{T}$, $\text{inc}^{-1}(U) = U \cap Y \in \mathcal{T}'$ so $\mathcal{T}' \supseteq \mathcal{T}|_Y$. $\square$

One can also define the subspace topology as the coarsest topology on Y such that inc is continuous, and derive the definition this way. Another point of view is that a function $f : (Z, \mathcal{T}_Z) \rightarrow (Y, \mathcal{T}|_Y)$ is continuous if and only if $\text{inc} \circ f : (Z, \mathcal{T}_Z) \rightarrow (X, \mathcal{T})$ is continuous.

Lemma 1.3 – Gluing lemma

Let $f : X \rightarrow Y$ be a function between topological spaces.

- (i) If $\{U_\alpha\}_{\alpha \in I}$ are open subsets which cover X and each $f|_{U_\alpha} : U_\alpha \rightarrow Y$ is continuous where U_α is given the subspace topology, then f is continuous.
- (ii) If $\{C_\alpha\}_{\alpha \in I}$ is a finite collection of closed sets covering X and each $f|_{C_\alpha} : C_\alpha \rightarrow Y$ is continuous, then f is continuous.

Proof. Let $V \subseteq Y$ be open. We want to show that $f^{-1}(V)$ is open.

$$f^{-1}(V) = f^{-1}(V) \cap X = f^{-1}(V) \cap \left(\bigcup_{\alpha \in I} U_\alpha \right) = \bigcup_{\alpha \in I} \underbrace{f^{-1}(V) \cap U_\alpha}_{= f|_{U_\alpha}^{-1}(V)}.$$

As $f|_{U_\alpha}$ is continuous, $f|_{U_\alpha}^{-1}(V)$ are open in U_α with the subspace topology. That is, there exists W open in X such that $f|_{U_\alpha}^{-1}(V) = U_\alpha \cap W$. This is an intersection of open subsets of X so it is open in X . Hence, $\bigcup_{\alpha \in I} f|_{U_\alpha}^{-1}(V)$ is a union of open sets, so it is open. The proof for (ii) is the same but we use the closed set definition of continuity. $\square$

Example 1.13

Suppose (X, d) is a metric space and $\mathcal{T}_d$ is its topology. Let $Y \subseteq X$. Then, $\mathcal{T}_d|_Y$ is the topology induced by $d|_Y$.

Example 1.14

$S^n = \{\mathbf{v} \in \mathbb{R}^{n+1} \mid |\mathbf{v}| = 1\}$ is the n -dimensional sphere. $D^{n+1} = \{\mathbf{v} \in \mathbb{R}^{n+1} \mid |\mathbf{v}| \leq 1\}$ is the $(n+1)$ -dimensional disc. $S^n \subset D^{n+1}$ and their metric topologies coincide with the subspace topology for $\mathbb{R}^{n+1}$.

Definition 1.11 – Quotient topology

Let $(X, \mathcal{T}_X)$ be a topological space and $\sim$ be an equivalence relation on X . $X/\sim$ are the equivalence classes and $\pi : X \rightarrow X/\sim$ is the quotient map. The quotient topology on $X/\sim$ is

$$\mathcal{T}_{X/\sim} = \{U \subseteq X/\sim \mid \pi^{-1}(U) \in \mathcal{T}_X\}.$$

Proposition 1.11

The quotient topology is a topology.

Proof. $\emptyset = \pi^{-1}(\emptyset) \in \mathcal{T}_X$ so $\emptyset \in \mathcal{T}_{X/\sim}$. $X = \pi^{-1}(X/\sim) \in \mathcal{T}_X$ so $X/\sim \in \mathcal{T}_{X/\sim}$. Let $\{U_\alpha\}_{\alpha \in I}$ be a collection of elements of $\mathcal{T}_{X/\sim}$, then $\pi^{-1}(\bigcup_{\alpha \in I} U_\alpha) = \bigcup_{\alpha \in I} \pi^{-1}(U_\alpha)$ and $\pi^{-1}(U_\alpha) \in \mathcal{T}_X$ so the union is too, hence $\bigcup_{\alpha \in I} U_\alpha \in \mathcal{T}_{X/\sim}$. Similar for finite intersections. $\square$

Proposition 1.12

The quotient map $\pi : (X, \mathcal{T}_X) \rightarrow (X/\sim, \mathcal{T}_{X/\sim})$ is continuous, and $\mathcal{T}_{X/\sim}$ is the finest topology for which this is true.

Proof. This is a tautology. $\square$

Observe that this is completely dual to the subspace topology. An alternative characterisation of the quotient topology is the following. For any space $(Z, \mathcal{T}_Z)$, a function $f : (X/\sim, \mathcal{T}_{X/\sim}) \rightarrow (Z, \mathcal{T}_Z)$ is continuous if and only if $f \circ \pi : (X, \mathcal{T}_X) \rightarrow (Z, \mathcal{T}_Z)$ is continuous.

Forgetting topology, a function $f : X/\sim \rightarrow Z$ is equivalent to giving a function $\tilde{f} : X \rightarrow Z$ such that $\tilde{f}(x) = \tilde{f}(x')$ when $x \sim x'$. Analogously, giving a continuous $f : (X/\sim, \mathcal{T}_{X/\sim}) \rightarrow (Z, \mathcal{T}_Z)$ is equivalent to giving a continuous $\tilde{f} : (X, \mathcal{T}_X) \rightarrow (Z, \mathcal{T}_Z)$ such that $\tilde{f}(x) = \tilde{f}(x')$ when $x \sim x'$.

Definition 1.12 – Quotient map

We say that a continuous function $g : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ is a quotient map if it is surjective and $U \in \mathcal{T}_Y$ if and only if $g^{-1}(U) \in \mathcal{T}_X$.

Given a quotient map g , consider an equivalence relation $\sim$ on X by $x \sim x'$ if and only if $g(x) = g(x')$. Then there is an induced function

$$\begin{aligned} G : X/\sim &\rightarrow Y \\ [x] &\mapsto g(x). \end{aligned}$$

This is a bijection. It is continuous by the universal property of quotients. Its inverse is also continuous so it is a homeomorphism.

Example 1.15

On $\mathbb{R}$, consider the equivalence relation $x \sim y$ if and only if $x - y \in \mathbb{Z}$. What is $\mathbb{R}/\sim$? Consider

$$\begin{aligned} f : \mathbb{R} &\rightarrow \mathbb{R}^2 \\ x &\mapsto (\sin(2\pi x), \cos(2\pi x)). \end{aligned}$$

This is continuous. It lands in $S^1 \subset \mathbb{R}^2$ so $f : \mathbb{R} \rightarrow S^1$ is continuous too by the universal property of the subspace topology. Notice that $x \sim y$ if and only if $f(x) = f(y)$. So we get a continuous bijection $F : \mathbb{R}/\sim \rightarrow S^1$. This is a homeomorphism.

Example 1.16

Consider $X = \mathbb{R} \times \{0, 1\} \subset \mathbb{R}^2$ with the subspace topology. Let $(x, i) \sim (y, j)$ if and only if $(x, i) = (y, j)$ or $x = y \neq 0$. Consider $X/\sim$. Any neighbourhood of $[(0, 0)]$ intersects any neighbourhood of $[(0, 1)]$, so it is not Hausdorff and so is not a subspace of $\mathbb{R}^2$.

For sets X and Y , $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$. There are projection functions

$$\begin{aligned} \pi_X : X \times Y &\rightarrow X & \pi_Y : X \times Y &\rightarrow Y \\ (x, y) &\mapsto x & (x, y) &\mapsto y. \end{aligned}$$

Definition 1.13 – Product topology

Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. The product topology $\mathcal{T}_{X \times Y}$ on $X \times Y$ consists of those $U \subseteq X \times Y$ such that for each $(x, y) \in U$, there is a $V \in \mathcal{T}_X$ and $W \in \mathcal{T}_Y$ such that $(x, y) \in V \times W \subseteq U$.

Proposition 1.13

The product topology is a topology.

Proof. $\emptyset \in \mathcal{T}_{X \times Y}$ tautologically. $X \times Y \in \mathcal{T}_{X \times Y}$ since we can take $V = X$ and $W = Y$. For a collection $\{U_\alpha\}_{\alpha \in I}$ of elements of $\mathcal{T}_{X \times Y}$, let $(x, y) \in \bigcup_{\alpha \in I} U_\alpha$. Then, $(x, y) \in U_\beta$ for some β . This is open so there exist $(x, y) \in V \times W \subseteq U_\beta \subseteq \bigcup_{\alpha \in I} U_\alpha$. If I is finite and $(x, y) \in \bigcap_{\alpha \in I} U_\alpha$, then $(x, y) \in V_\alpha \times W_\alpha \subseteq U_\alpha$ for each $\alpha \in I$. So, $(x, y) \in (\bigcap_{\alpha \in I} V_\alpha) \times (\bigcap_{\alpha \in I} W_\alpha) \subseteq \bigcap_{\alpha \in I} U_\alpha$ and these intersections are finite so $\bigcap_{\alpha \in I} V_\alpha$ and $\bigcap_{\alpha \in I} W_\alpha$ are open. $\square$

Since it is a topology, we see that the set $V \times W$ are a basis for it.

Proposition 1.14

The projection maps $\pi_X : (X \times Y, \mathcal{T}_{X \times Y}) \rightarrow (X, \mathcal{T}_X)$ and $\pi_Y : (X \times Y, \mathcal{T}_{X \times Y}) \rightarrow (Y, \mathcal{T}_Y)$ are continuous, and $\mathcal{T}_{X \times Y}$ is the coarsest topology for this is true.

Proof. Let $V \in \mathcal{T}_X$. $\pi_X^{-1}(V) = V \times Y$ which is open. So, π_X and similarly π_Y is continuous. If $\mathcal{T}'$ is a topology on $X \times Y$ such that π_X and π_Y are continuous, then $\pi_X^{-1}(V) = V \times Y$ and $\pi_Y^{-1}(W) = X \times W$ is open. Their intersection $V \times W$ is open in $\mathcal{T}'$. Hence, $\mathcal{T}_{X \times Y} \subseteq \mathcal{T}'$. $\square$

The universal property of the product topology is the following. A function $f : (Z, \mathcal{T}_Z) \rightarrow (X \times Y, \mathcal{T}_{X \times Y})$ is continuous if and only if $\pi_X \circ f : (Z, \mathcal{T}_Z) \rightarrow (X, \mathcal{T}_X)$ and $\pi_Y \circ f : (Z, \mathcal{T}_Z) \rightarrow (Y, \mathcal{T}_Y)$ are both continuous. Equivalently, writing $f = (f_1, f_2)$ with $f_1 : Z \rightarrow X$ and $f_2 : Z \rightarrow Y$, then f is continuous if and only if f_1 and f_2 are continuous.

2 Connectedness and components

The intermediate value theorem states that if $f : [0, 1] \rightarrow \mathbb{R}$ is continuous with $f(0) < 0 < f(1)$, then $f(t) = 0$ for some $t \in (0, 1)$. This is a statement about continuous functions but is also about the interval $[0, 1]$. For example, $f : [0, 1/2) \cup (1/2, 1] \rightarrow \mathbb{R}$ does not satisfy the condition.

Definition 2.1 – Connectedness

A topological space X is disconnected if $X = U \cup V$ where U and V are disjoint non-empty open sets. Otherwise, X is connected.

If $X = U \cup V$ is disconnected, then U and V are open and closed.

Example 2.1

Any set with the coarse topology is connected. Any set with the discrete topology is disconnected if it has more than one point.

Example 2.2

$X = [0, 1/2) \cup (1/2, 1] \subseteq \mathbb{R}$ is disconnected since $[0, 1/2) = X \cap (-\infty, 1/2)$ and $(1/2, 1] = X \cap (1/2, \infty)$ are both open in X .

Proposition 2.1

A space X is disconnected if and only if there is a continuous surjection $f : X \rightarrow \{0, 1\}$ where $\{0, 1\}$ has the discrete topology.

Proof. Suppose $X = U \cup V$ is disconnected. Let $f : X \rightarrow \{0, 1\}$ be defined by

$$f(x) = \begin{cases} 0 & \text{if } x \in U \\ 1 & \text{if } x \in V. \end{cases}$$

This is well-defined since U and V are disjoint. This is surjective since U and V are non-empty. f is continuous as the open sets in $\{0, 1\}$ are $\emptyset, \{0\}, \{1\}, \{0, 1\}$ and their preimages are $\emptyset, U, V, X$ which are all open.

Conversely, given such a $f : X \rightarrow \{0, 1\}$, let $U = f^{-1}(0)$ and $V = f^{-1}(1)$. They are disjoint and non-empty since f is surjective. They are open since f is continuous and $\{0\}, \{1\}$ are open. $\square$

Theorem 2.1

The spaces $[0, 1], [0, 1), (0, 1)$ are all connected.

Proof. We only consider $[0, 1]$, the other cases are similar. If it is disconnected, there is a continuous surjection $f : [0, 1] \rightarrow \{0, 1\} \subseteq \mathbb{R}$. Then, $f(\cdot) - 1/2 : [0, 1] \rightarrow \mathbb{R}$ is continuous and takes the values $\pm 1/2$. The intermediate value theorem implies that it takes the value 0. This is a contradiction as it only takes the values $\pm 1/2$. $\square$

Theorem 2.2 – Intermediate value theorem

Let X be a topological space that is connected and $f : X \rightarrow \mathbb{R}$ be a continuous function. If there are $x_0, x_1 \in X$ such that $f(x_0) < 0 < f(x_1)$, then there is $x_2 \in X$ such that $f(x_2) = 0$.

Proof. Consider the open sets $U = f^{-1}((-\infty, 0))$ and $V = f^{-1}((0, \infty))$. They are open as they are the preimage of open sets. They are non-empty as $x_0 \in U$ and $x_1 \in V$. They are disjoint. If $f(x) \neq 0$ for all $x \in X$, then $X = U \cup V$ and X is disconnected. But this is not the case, so there must be some $x_2 \in X$ with $f(x_2) = 0$. $\square$

Proposition 2.2

Let $f : X \rightarrow Y$ be a continuous surjection. If X is connected, then Y is connected.

Proof. We prove the contrapositive. If Y is disconnected, then we have a continuous surjection $h : Y \rightarrow \{0, 1\}$. Hence, $h \circ f : X \rightarrow \{0, 1\}$ is continuous and surjective so X is disconnected. $\square$

Corollary 2.1

If X is connected and $f : X \rightarrow Y$ is continuous, then $\text{im}(f)$ is connected.

Proof. Apply Proposition 2.2 to $f : X \rightarrow \text{im}(f) \subseteq Y$ which is surjective by definition. $\square$

Example 2.3

If X is a connected space and $\sim$ is an equivalence relation, then $\pi : X \rightarrow X/\sim$ is continuous and surjective, so the quotient space $X/\sim$ is connected.

Lemma 2.1

If $f : X \rightarrow Y$ is a homeomorphism and $Z \subseteq X$ is a subspace, then $f|_Z : Z \rightarrow f(Z)$ is a homeomorphism.

Proof. $f^{-1}|_{f(Z)}$ gives a continuous inverse. $\square$

Example 2.4

$[0, 1]$ is not homeomorphic to $(0, 1)$. If $f : [0, 1] \rightarrow (0, 1)$ were a homeomorphism, it gives a homeomorphism $f|_{(0, 1]} : (0, 1] \rightarrow (0, 1) \setminus \{f(0)\} = X$. $(0, 1]$ is connected by Theorem 2.1. $X = (0, f(0)) \cup (f(0), 1)$ since $0 < f(0) < 1$, so X is not connected as they are disjoint non-empty open subsets. This is a contradiction by Proposition 2.2. Hence, the two spaces are not homeomorphic. Similarly, $[0, 1]$ is not homeomorphic to either $[0, 1]$ or $(0, 1)$.

Example 2.5

S^1 is not homeomorphic to $\mathbb{R}$. Recall that $S^1 \cong \mathbb{R}/\sim$ where $x \sim y$ if and only if $x - y \in \mathbb{Z}$. We apply the same trick as in Example 2.4. $S^1 \setminus \{1\}$ is homeomorphic to $(0, 1)$ via $t \mapsto e^{2\pi it}$ so it is connected. But if we remove any point from $\mathbb{R}$ then it becomes disconnected.

Proposition 2.3

Let $\{X_\alpha\}_{\alpha \in I}$ be a collection of subspaces of X . Suppose each of X_α is connected and $\bigcap_{\alpha \in I} X_\alpha \neq \emptyset$. Then, $\bigcup_{\alpha \in I} X_\alpha$ is connected.

Proof. Suppose $X = \bigcup_{\alpha \in I} X_\alpha$. If it is disconnected with $X = U \cup V$, then $U \cap X_\alpha$ and $V \cap X_\alpha$ are disjoint open sets covering X_α . As X_α is connected, one of two has to be empty, so $X_\alpha \subseteq U$ or $X_\alpha \subseteq V$. This holds for each X_α . As $\bigcap_{\alpha \in I} X_\alpha \neq \emptyset$, the X_α 's are all in U or all in V . Hence, $\bigcup_{\alpha \in I} X_\alpha$ is U or V , so V or U is empty. This is a contradiction so X is connected. $\square$

Corollary 2.2

If X and Y are connected, then $X \times Y$ is connected.

Proof. We can suppose X and Y are both non-empty since $\emptyset$ is connected. Choose $x \in X$. Consider $C_y = \{x\} \times Y \cup X \times \{y\}$. The sets $\{x\} \times Y \cong Y$ and $X \times \{y\} \cong X$ intersect at (x, y) . They are connected by assumption so C_y is connected by Proposition 2.3. Observe that $X \times Y = \bigcup_{y \in Y} C_y$ and $\bigcap_{y \in Y} C_y = \{x\} \times Y \neq \emptyset$. Using Proposition 2.3 again, $X \times Y$ is connected. $\square$

Definition 2.2 – Path and path-connected

If X is a space and $x_0, x_1 \in X$, a path between them is a continuous function $\gamma : [0, 1] \rightarrow X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$. We say that X is path-connected if for any two points in X , there is a path between them.

Example 2.6

The spaces $X \in \{(0, 1), (0, 1], [0, 1], \mathbb{R}\}$ are all path-connected. For any $x_0, x_1 \in X$, consider $\gamma(t) = (1 - t)x_0 + tx_1$. This is continuous. For the same reason, $\mathbb{R}^n$ is path-connected, and any convex subspace $X \subseteq \mathbb{R}^n$ is also path-connected tautologically.

Example 2.7

$X = \mathbb{R}^2 \setminus \{0\}$ is path-connected. For points that do not lie on a line through the origin, we can do linear interpolation as in Example 2.6. If they do, use a bent path.

Proposition 2.4

A path-connected space X is connected.

Proof. If X is not connected, we have a continuous surjective function $f : X \rightarrow \{0, 1\}$. Let $x_0, x_1 \in X$ be such that $f(x_0) = 0$ and $f(x_1) = 1$. Suppose X is path-connected, then there is a path $\gamma : [0, 1] \rightarrow X$ joining x_0 to x_1 . So, $f \circ \gamma : [0, 1] \rightarrow \{0, 1\}$ is continuous and surjective. This shows that $[0, 1]$ is connected which is a contradiction. $\square$

Example 2.8

$\mathbb{R}^n \not\cong \mathbb{R}$ for $n > 1$. If it were, then $\mathbb{R}^n \setminus \{0\} \cong \mathbb{R} \setminus \{t\}$. The right hand side is disconnected. However, the left hand side is path-connected and hence is connected by Proposition 2.4. Contradiction.

Let X be a space. Define $\sim$ on X by $x \sim y$ if and only if there exists a path $\gamma : [0, 1] \rightarrow X$ from x to y .

Lemma 2.2

$\sim$ is an equivalence relation.

Proof. $x \sim x$ since $\gamma(t) = x$ is a path from x to itself. If $x \sim y$, there exists γ from x to y . Then, $\gamma(1 - t)$ is a path from y to x , so $y \sim x$. If $x \sim y$ and $y \sim z$, then there are paths γ from x to y and γ' from y to z .

$$(\gamma' \cdot \gamma)(t) = \begin{cases} \gamma(2t) & \text{if } 0 \leq t \leq 1/2 \\ \gamma'(2t - 1) & \text{if } 1/2 \leq t \leq 1 \end{cases}$$

is a path from x to z . It is continuous by the gluing lemma applied to $[0, 1/2], [1/2, 1] \subseteq [0, 1]$. So, $x \sim z$. $\square$

Definition 2.3 – Path components

The path components of X are the equivalence classes of $\sim$.

Each path component is path-connected. If x and y are in the same path component, there exists a path $\gamma : [0, 1] \rightarrow X$ between them in X . As $\gamma|_{[0, t]} : [0, t] \rightarrow X$ is a path from $x = \gamma(0)$ to $\gamma(t)$, this path lies wholly in a path component of X .

Let X be a space. Define $\approx$ on X by $x \approx y$ if and only if there exists a connected subspace $C \subseteq X$ with $x, y \in C$.

Lemma 2.3

$\approx$ is an equivalence relation.

Proof. $\{x\}$ is connected so $x \approx x$. We have $x \approx y \iff y \approx x$ by definition. If $x \approx y$ and $y \approx z$, then there exist connected $C_1, C_2 \subseteq X$ such that $x, y \in C_1$ and $y, z \in C_2$. Since $C_1 \cap C_2 \supseteq \{y\}$, $C_1 \cup C_2$ is connected by Proposition 2.3. $x, z \in C_1 \cup C_2$ so $x \approx z$. $\square$

Definition 2.4 – Connected components

The connected components of X are the equivalence classes of $\approx$.

Proposition 2.5

Connected components are connected.

Proof. Let $C \subseteq X$ be a connected component. Suppose $f : C \rightarrow \{0, 1\}$ is a continuous surjective function. Let $f(x_0) = 0$ and $f(x_1) = 1$. As $x_0 \approx x_1$, there exists a connected $D \subseteq X$ with $x_0, x_1 \in D$. If $d \in D$, then $d \approx x_0$, so $D \subseteq C$. But then $f|_D : D \rightarrow \{0, 1\}$ is continuous and surjective, contradicting that D is connected. $\square$

Example 2.9

$X = (-\infty, 0) \cup (0, \infty) \subset \mathbb{R}$ has connected components $(-\infty, 0)$ and $(0, \infty)$, which are also path-connected.

Example 2.10

The connected components of $\mathbb{Q} \subset \mathbb{R}$ are the singletons $\{x\}$, which are also the path-components.

Example 2.11 – Topologist's sine curve

Consider $S = \{(x, \sin(1/x)) \in \mathbb{R}^2 \mid 0 < x \leq 1\}$. This is not closed since we can construct a sequence converging to say $(0, 1)$. Its closure is $\bar{S} = S \cup (\{0\} \times [-1, 1])$. We want to show that $\bar{S}$ is connected but not path-connected. As S is the image of the connected space $(0, 1]$ under a continuous function, it is

connected. By the following Lemma 2.4, $\bar{S}$ is connected.

Lemma 2.4

The closure of a connected subspace is connected.

Proof. Let $C \subseteq X$ be a connected subspace. Suppose $\bar{C} = U \cup V$ is disconnected, so U and V are non-empty disjoint open subsets of $\bar{C}$. Then, U and V are also closed in $\bar{C}$, so are closed in X too. $C \cap U$ and $C \cap V$ are two disjoint open subsets of C . Since C is connected, one of the intersections is empty, say $C \cap U = \emptyset$. As U is closed in X , we must have $\bar{C} \subset U$ too. Then, $\bar{C} \cap V = \emptyset$. This contradicts U and V disconnecting $\bar{C}$. $\square$

Note that S and $\{0\} \times [-1, 1]$ are both path-connected, so to show that $\bar{S}$ is not path-connected, we need to show that there is not path $\gamma : [0, 1] \rightarrow \bar{S}$ with $\gamma(0) \in \{0\} \times [-1, 1]$ and $\gamma(1) \in S$. Suppose γ is such a path. As $\{0\} \times [-1, 1]$ is closed, $\gamma^{-1}(\{0\} \times [-1, 1])$ is closed, so it contains its supremum t . Then, $\gamma|_{[t, 1]} : [t, 1] \rightarrow \bar{S}$ is a path with $\gamma(t) \in \{0\} \times [-1, 1]$ and $\gamma(s) \in S$ for all $s > t$. We can replace γ with $\gamma|_{[t, 1]}$. Write $\gamma(s) = (x(s), y(s))$ for $x, y : [0, 1] \rightarrow \mathbb{R}$ continuous. For $s > 0$, $y(s) = \sin(1/x(s))$.

Our goal is to find a sequence (s_n) with $s_n \rightarrow 0$ and $y(s_n) = (-1)^n$. This provides a contradiction since it should converge to $y(0)$ but it does not. For each n , choose $0 < u < x(1/n)$ such that $\sin(1/u) = (-1)^n$. By the intermediate value theorem, we have $u = x(s_n)$ for some $0 < s_n < 1/n$. Then, $y(s_n) = \sin(1/x(s_n)) = \sin(1/u) = (-1)^n$ as required.

3 Compactness

Definition 3.1 – Cover, open cover and subcover

A collection $\mathcal{X} \in \mathcal{P}(X)$ is a cover of X if for each $x \in X$, there is a $S \in \mathcal{X}$ with $x \in S$. An open cover of X is a cover consisting of open sets. A subcover of $\mathcal{X}$ is a subset $\mathcal{X}' \subseteq \mathcal{X}$ which is also a cover.

Definition 3.2 – Compact

A topological space X is compact if every open cover $\mathcal{U}$ of X has a finite subcover $\mathcal{U}' \subseteq \mathcal{U}$.

Example 3.1

$\mathbb{R}$ is not compact. Consider the open sets $\{(n-1, n+1) \subset \mathbb{R} \mid n \in \mathbb{Z}\}$. This covers $\mathbb{R}$. However, there is no proper subcover as $m \in \mathbb{Z}$ lies $(n-1, n+1)$ if and only if $m = n$.

Example 3.2

If X is a topological space with finitely many points, then it is compact. Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in I}$ be an open cover. Each $x \in X$ lies in some U_α and call it U_x . Then, $\mathcal{U}' = \{U_x\}_{x \in X}$ is a finite subcover.

Example 3.3

$X = \{0\} \cup \{1/n \mid n \in \mathbb{Z}^+\} \subset \mathbb{R}$ is compact. Let $\mathcal{U}$ be an open cover of X . The point $0 \in X$ lies in some $U_0 \in \mathcal{U}$. As this is open, there exists $\varepsilon > 0$ such that $0 \in B_\varepsilon(0) = (-\varepsilon, \varepsilon) \subseteq U_0$. Then, $1/n \in U_0$ for all $n > 1/\varepsilon$. This leaves finitely many points $1, 1/2, \dots, 1/n$ that lie in open sets with $1 \in U_1, 1/2 \in U_2, \dots, 1/n \in U_n$. Then, $\{U_0, U_1, \dots, U_n\}$ is a finite subcover.

The space $Y = \{1/n \mid n \in \mathbb{Z}^+\}$ is not compact. This is because it is an infinite set and the topology is discrete. $\{1/n\} = Y \cap (1/(n+1), 1/(n-1))$. The open cover $\{1/n\}_{n \in \mathbb{Z}^+}$ has no finite subcover.

Theorem 3.1

$[0, 1]$ is compact.

Proof. Let $\mathcal{U}$ be an open cover of $[0, 1]$. Consider

$$A = \{a \in [0, 1] \mid \text{there is a finite } \mathcal{U}' \subseteq \mathcal{U} \text{ whose union contains } [0, a]\}.$$

If $0 \leq a \leq b$, then $[0, a] \subseteq [0, b]$. So, if $b \in A$, then $a \in A$. Also, $0 \in A$ since $[0, 0]$ is just the singleton $\{0\}$. It therefore has a supremum $\alpha \in [0, 1]$. We want to show that $\alpha = 1$. Let $\alpha \in U_\alpha \in \mathcal{U}$. If $\alpha < 1$, then there is a $\varepsilon > 0$ such that $(\alpha - \varepsilon, \alpha + \varepsilon) \in U_\alpha$. Also, there exists $a \in A$ such that $\alpha - \varepsilon < a \leq \alpha$. So, $\alpha - \varepsilon \in A$ too. Then, $[0, \alpha - \varepsilon]$ has a finite cover by elements of $\mathcal{U}$. Adding U_α to this collection, we obtain a finite list of elements of $\mathcal{U}$ that cover $[0, \alpha + \varepsilon/2]$. Then, $\alpha + \varepsilon/2 \in A$. This contradicts our definition of the supremum. Hence, $\alpha = 1$. $\square$

Proposition 3.1

If X is compact and $C \subseteq X$ is closed, then C is compact.

Proof. Open sets in C are of the form $C \cap U_\alpha$ where U_α is open in X . Suppose $\{C \cap U_\alpha \mid \alpha \in I\}$ is an open cover. Then, $(X \setminus C) \cup \{U_\alpha \mid \alpha \in I\}$ is an open cover of X since $C = \bigcup_{\alpha \in I} C \cap U_\alpha \subseteq \bigcup_{\alpha \in I} U_\alpha$. As X is compact, this has a finite subcover. It can be taken to have the form $(X \setminus C) \cup \{U_\alpha \mid \alpha \in I'\}$ such that $I' \subseteq I$ is finite. As these cover X , their intersections with C cover C . Hence, $\{C \cap U_\alpha \mid \alpha \in I'\}$ is a finite subcover. $\square$

Proposition 3.2

If X is Hausdorff and $C \subseteq X$ is compact, then C is closed.

Proof. We will show that $U = X \setminus C$ is open, so for every $x \in U$, we need to find an open neighbourhood $U_x \subseteq U$ of x . Fix $x \in U$. For each $y \in C$, we can use Hausdorffness to find disjoint open sets $V_y \ni y$ and $W_y \ni x$. Then, $\{C \cap V_y \mid y \in C\}$ is an open cover of C . Since C is compact, we can find $y_1, \dots, y_n \in C$ such that $\{C \cap V_{y_i} \mid i = 1, \dots, n\}$ cover C . Let $U_x = \bigcap_{i=1}^n W_{y_i}$. This contains x and is a finite intersection of open sets, so it is an open neighbourhood of x . As W_{y_i} is disjoint from V_{y_i} for each i , $U_x = \bigcap_{i=1}^n W_{y_i}$ is disjoint from $\bigcap_{i=1}^n V_{y_i} \supseteq C$. So, U_x is disjoint from C and $U_x \subseteq U$. Doing this for every $x \in U$ shows that U is open. $\square$

Proposition 3.3

If X is compact and $f : X \rightarrow Y$ is continuous, then $f(X)$ is a compact subspace.

Proof. We can suppose f is surjective by replacing Y with $f(X)$. If $\mathcal{U}$ is an open cover of Y , consider $f^{-1}(\mathcal{U}) = \{f^{-1}(U) \mid U \in \mathcal{U}\}$. Since f is continuous, $f^{-1}(U)$ is open so this is an open cover of X . By compactness, it has a finite subcover $f^{-1}(U_1), \dots, f^{-1}(U_n)$. Since $X = \bigcup_{i=1}^n f^{-1}(U_i)$, we have $Y = f(X) = \bigcup_{i=1}^n U_i$. Hence, $U_1, \dots, U_n$ is a finite subcover. $\square$

Corollary 3.1

If $f : X \rightarrow Y$ is a continuous bijection from a compact space X to a Hausdorff space Y , then it is a homeomorphism.

Proof. We want to show that $f^{-1} : Y \rightarrow X$ is continuous. We use the closed set characterisation of continuity in this proof. Let $C \subseteq X$ be closed, we want $(f^{-1})^{-1}(C) = f(C)$ to be closed. Since X is compact, C is compact by Proposition 3.1. By Proposition 3.3, $f(C)$ is compact. By Proposition 3.2, $f(C)$ is closed. $\square$

Recall that a metric space (X, d) is sequentially compact if every sequence in X has a convergent subsequence. The metric d gives a topology $\mathcal{T}_d$ on X . We want to compare sequential compactness and compactness.

Lemma 3.1 – Lebesgue number lemma

Let (X, d) be sequentially compact and $\mathcal{U} \subseteq \mathcal{T}_d$ be an open cover. Then, there is a $\delta > 0$ such that each $B_\delta(x)$ lies inside some element of $\mathcal{U}$.

Proof. Suppose not. Then, for each $n \in \mathbb{Z}^+$, there exists a $x_n \in X$ such that $B_{1/n}(x_n)$ does not lie in any element of $\mathcal{U}$. By sequential compactness, there is a convergent subsequence $x_{n_i} \rightarrow x_\infty$. Let $x_\infty \in U \in \mathcal{U}$ and $\varepsilon > 0$ be such that $B_\varepsilon(x_\infty) \subseteq U$. For all sufficiently large i , we have $x_{n_i} \in B_{\varepsilon/2}(x_\infty)$ and $1/n_i < \varepsilon/2$. By the triangle inequality, $B_{1/n_i}(x_{n_i}) \subseteq B_\varepsilon(x_\infty) \subseteq U$, hence a contradiction. $\square$

Note that such a $\delta > 0$ is called a Lebesgue number for the open cover $\mathcal{U}$.

Theorem 3.2

A metric space (X, d) is sequentially compact if and only if the topological space $(X, \mathcal{T}_d)$ is compact.

Proof. Suppose (X, d) is not sequentially compact. Then, there is a sequence (t_n) in X with no convergent subsequence. That is, for each $x \in X$, there is an open neighbourhood $U_x \ni x$ which only contains finitely many terms. Then, $\mathcal{U} = \{U_x \mid x \in X\}$ is an open cover of $(X, \mathcal{T}_d)$. If it had a finite subcover, then as each U_x contains only finitely many terms, it follows that there are only finitely many terms. This is obviously impossible so $\mathcal{U}$ does not have a finite subcover. Hence, $(X, \mathcal{T}_d)$ is not compact.

Now suppose that (X, d) is sequentially compact. Let $\mathcal{U}$ be an open cover of $(X, \mathcal{T}_d)$. Let $\delta > 0$ be a Lebesgue number for $\mathcal{U}$, which exists by Lemma 3.1. We want to show that there is a finite $A \subseteq X$ such that $X = \bigcup_{a \in A} B_\delta(a)$. If not, then for every finite set A , there exists $x \in X$ such that $d(x, a) > \delta$ for all $a \in A$. So, we can form a sequence $x_1, x_2, \dots$ such that $d(x_i, x_j) \geq \delta$ for all $i \neq j$. This property means it has no convergent subsequence, a contradiction. So, such a finite set A does exist. Each $B_\delta(a)$ lie in some $U_a \in \mathcal{U}$ since δ is a Lebesgue number. Hence, $\{U_a \mid a \in A\}$ is a finite subcover of $\mathcal{U}$, and $(X, \mathcal{T}_d)$ is compact. $\square$

Corollary 3.2 – Heine-Borel

A subspace $X \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded.

Proof. In Analysis II, we proved this with compactness replaced with sequential compactness. Theorem 3.2 says that the two notions are the same. $\square$

Corollary 3.3 – Extreme value theorem

If X is a compact topological space and $f : X \rightarrow \mathbb{R}$ is continuous, then there exists $a, b \in X$ such that $f(a) \leq f(x) \leq f(b)$ for all $x \in X$.

Proof. The set $f(X) \subseteq \mathbb{R}$ is compact by Proposition 3.3. By Heine-Borel, it is closed and bounded. So, it contains its supremum $A = f(a)$ and its infimum $B = f(b)$. Hence, $f(X) \subseteq [A, B]$ and we are done. $\square$

Theorem 3.3

If X and Y are compact, then the product space $X \times Y$ is compact.

Proof. We first prove the following, which only uses that Y is compact.

Lemma 3.2 – Tube lemma

If $x_0 \in X$ and $\{x_0\} \times Y \subseteq W$ is an open neighbourhood, then there is an open neighbourhood $U_{x_0} \ni x_0$ such that $U_{x_0} \times Y \subseteq W$.

Proof. We can cover $\{x_0\} \times Y$ by open sets of the form $U \times V \subseteq W$ where U and V are open. We can do this since W is open and sets of the form $U \times V$ are a basis by the definition of product topology. As $\{x_0\} \times Y$ is compact, this cover has a finite subcover, so there are open sets $U_1 \times V_1, \dots, U_n \times V_n \subseteq W$ that cover $\{x_0\} \times Y$. Let $U_{x_0} = \bigcap_{i=1}^n U_i$ which is an open neighbourhood of x_0 . To see that $U_{x_0} \times Y \subseteq W$, consider $(x, y) \in U_{x_0} \times Y$. Note that $(x_0, y) \in \{x_0\} \times Y$ so it lies in some $U_i \times V_i$. So, $y \in V_i$. Also, $x \in U_{x_0} \subseteq U_i$. Thus, $(x, y) \in U_i \times V_i \subseteq W$. $\square$

To show that $X \times Y$ is compact, let $\mathcal{W}$ be an open cover of $X \times Y$. For each $x_0 \in X$, $\{x_0\} \times Y$ can be covered by finitely many $W_1, \dots, W_n \in \mathcal{W}$ since it is compact. So, $\{x_0\} \times Y \subseteq \bigcup_{i=1}^n W_i$. By the tube lemma, there is a neighbourhood $U_{x_0} \ni x_0$ such that $U_{x_0} \times Y \subseteq \bigcup_{i=1}^n W_i$. That is, $U_{x_0} \times Y$ can be covered by finitely many elements of $\mathcal{W}$. Doing this for every point, we obtain $\{U_{x_0} \mid x_0 \in X\}$ which is an open cover of X . This has a finite subcover $U_{x_0}, U_{x_1}, \dots, U_{x_m}$ since X is compact. Thus, $X \times Y$ is covered by $U_{x_0} \times Y, \dots, U_{x_m} \times Y$. As each of these can be covered by finitely many elements of $\mathcal{W}$, so can the whole of $X \times Y$. $\square$

Corollary 3.4

$[0, 1]^n$ is compact.

Proof. $[0, 1]$ is compact. Apply Theorem 3.3 inductively. $\square$

This gives us another proof of Heine-Borel without using analysis.

Proof. Let X be closed and bounded, so $X \subseteq [-N, N]^n \subset \mathbb{R}^n$ for some $N > 0$. By Corollary 3.4, $[-N, N]^n$ is compact. Since X is a closed subspace, it is compact by Proposition 3.1.

Conversely, if X is compact, then it is closed by Proposition 3.2 and the fact that $\mathbb{R}^n$ is Hausdorff. To see that it is bounded, consider the open cover $\{X \cap (-N, N)^n \mid N \in \mathbb{N}\}$. This must have a finite subcover, say $X \cap (-N_1, N_1)^n, \dots, X \cap (-N_k, N_k)^n$. If $N_1 \leq \dots \leq N_k$, then these are nested. So, $X = X \cap (-N_k, N_k)^n$ and hence $X \subseteq (-N_k, N_k)^n$. $\square$

Definition 3.3 – Finite intersection property

We say a collection $\{S_\alpha \mid \alpha \in I\}$ of subsets of a fixed set S has the finite intersection property if any finite intersection of S_α 's is non-empty.

Theorem 3.4

A space X is compact if and only if every collection of closed subsets $\{C_\alpha \mid \alpha \in I\}$ with the finite intersection property has $\bigcap_{\alpha \in I} C_\alpha \neq \emptyset$.

Proof. Suppose first that X is compact, and let $\mathcal{U} = \{X \setminus C_\alpha : \alpha \in I\}$. This is a collection of open subsets, and if we suppose that $\bigcap_{\alpha \in I} C_\alpha = \emptyset$ then $\mathcal{U}$ is an open cover, as every point of X lies outside of some C_α . By compactness it has a finite subcover, meaning that there is a finite list $X \setminus C_{\alpha_1}, \dots, X \setminus C_{\alpha_n}$ which cover X : equivalently $\bigcap_{i=1}^n C_{\alpha_i} = \emptyset$, contradicting the finite intersection property.

Conversely, if a space X has the property in the statement and $\mathcal{U} = \{U_\alpha : \alpha \in I\}$ is an open cover, then $\{X \setminus U_\alpha : \alpha \in I\}$ is a collection of closed sets whose intersection is empty. Thus this collection *cannot* have the finite intersection property, and so there is a finite list $X \setminus U_{\alpha_1}, \dots, X \setminus U_{\alpha_n}$ whose intersection is empty: equivalently, $\bigcup_{i=1}^n U_{\alpha_i} = X$, giving a finite subcover. $\square$

Define a subspace $C \subseteq [0, 1]$ as follows. Let $C_0 = [0, 1]$. Remove its open middle third so $C_1 = [0, 1/3] \cup [2/3, 1]$. Remove the open middle third of each of the segments again so $C_2 = [0, 1/9] \cup [2/9, 1/3] \cup [2/3, 7/9] \cup [8/9, 1]$. We carry on defining $C_3, C_4, \dots$. The cantor set is

$$C = \bigcap_{i=0}^{\infty} C_i \subset [0, 1].$$

Since all the C_i 's are closed, C is closed, hence C is compact. It contains all the endpoints of the intervals in C_i .

Proposition 3.4

There is a bijection between C and $\{0, 1\}^{\mathbb{N}}$, so C has the cardinality of $[0, 1]$.

Proof. Let $\{0, 1\} = \{l, r\}$ where l represents left and r represents right. A sequence of l 's and r 's tells us how to choose one of the intervals in each C_i . Write such a choice as $[0, 1] = I_0 \supset I_1 \supset I_2 \supset \dots$. Using Theorem 3.4 and the fact that $[0, 1]$ is compact, $\bigcap_{i=0}^{\infty} I_i \neq \emptyset$. Suppose $x, y \in \bigcap_{i=0}^{\infty} I_i$ with $x \neq y$, then $x, y \in I_i$ for all i . But $|x - y| > 1/3^i = |I_i|$ for some i , thus contradiction. So, $\bigcap_{i=0}^{\infty} I_i$ is a single point and it lies in C . This defines a function $\phi : \{l, r\}^{\mathbb{N}} \rightarrow C$. This is injective since sequences which differ at the i -th position map to points that are $\geq 1/3^i$ apart. This is tautologically surjective since points in C lie in a unique I_i for each i . $\square$

In fact, $\phi^{-1} : C \rightarrow \{l, r\}^{\mathbb{N}}$ is continuous, so it is a homeomorphism.

Proposition 3.5

The connected components of C are the one-point spaces.

Proof. If $x, y \in C$ with $x \neq y$, then there exists $z \notin C$ with $x < z < y$. If not, $[x, y] \in C \subseteq C_i$. But the lengths of the intervals in C_i are $1/3^i$, so this cannot hold for all i . Then, $C \cap (-\infty, z)$ and $C \cap (z, \infty)$ disconnects C with x in one space and y in the other. Hence, x and y cannot lie in a connected subspace. $\square$

We can show that any non-empty compact metric space has a continuous surjection from C . Hence, any non-empty compact metric space has at most the cardinality of $\mathbb{R}$.

4 Topology of manifolds

Definition 4.1 – Manifold

A topological space X is a d -dimensional manifold if it is Hausdorff and is locally Euclidean, that is, every $x \in X$ has an open neighbourhood homeomorphic to an open subset of $\mathbb{R}^d$.

Note that any open neighbourhood $y \in U \subseteq \mathbb{R}^d$ is contained in some ball $B_\varepsilon(y)$, which is homeomorphic to $\mathbb{R}^d$. Thus, every point in a topological manifold has a neighbourhood homeomorphic to $\mathbb{R}^d$. We call this a Euclidean neighbourhood.

Example 4.1

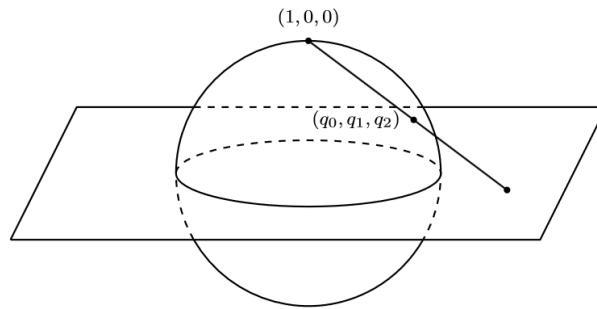
Any open $U \subseteq \mathbb{R}^d$ is a d -dimensional manifold.

Example 4.2 – Sphere

The sphere $S^d = \{v \in \mathbb{R}^{d+1} \mid \|v\| = 1\}$ is a manifold of dimension d . It is Hausdorff since it is a subspace of a Hausdorff space. To see it is locally Euclidean, let $p = (-1, 0, \dots, 0)$. Consider the stereographic projection

$$\begin{aligned} S^d \setminus \{-p\} &\rightarrow \mathbb{R}^d \\ q &\mapsto \text{the intersection of the line } (-p)q \text{ with the plane } \{0\} \times \mathbb{R}^d \\ (q_0, q_1, \dots, q_d) &\mapsto \frac{1}{1-q_0}(q_1, \dots, q_d) \\ \frac{1}{1+\sum e_i^2}(-1 + \sum e_i^2, 2e_1, \dots, 2e_d) &\leftarrow (e_1, \dots, e_d) \end{aligned}$$

which is a homeomorphism. This gives a Euclidean neighbourhood of any point except p , but we can do the same with p and $-p$ interchanged.



Example 4.3 – Circle

In particular, the circle S^1 is a 1-dimensional manifold. It can be described in other ways too. By Example 1.15, $S^1 \cong \mathbb{R}/\sim$ where $x \sim y$ iff $x - y \in \mathbb{Z}$. Note that any equivalence class of $\sim$ intersects $[0, 1]$, so using the continuous surjection $\pi \circ \text{inc} : [0, 1] \rightarrow \mathbb{R} \rightarrow \mathbb{R}/\sim$ shows that $\mathbb{R}/\sim$ is the continuous image of a compact space, so is compact. Similarly, $S^1 \cong [0, 1]/(0 \sim 1)$.

Example 4.4 – Line with two origins

Recall Example 1.16 with $L = (\mathbb{R} \times [0, 1]) / ((t, 0) \sim (t, 1) \text{ for all } t \neq 0)$. We have seen that it is not Hausdorff, so it is not a manifold. But it is locally Euclidean. Let π be the quotient map and $U = \pi((-1, 1) \times \{1\})$. Note that $\pi^{-1}(U) = (-1, 1) \times \{1\} \cup (-1, 0) \times \{0\} \cup (0, 1) \times \{0\}$ is open, so $U \subseteq L$ is open. Note $\pi|_{(-1, 1) \times \{1\}} : (-1, 1) \times \{1\} \rightarrow U$ is a homeomorphism. So, U is a Euclidean neighbourhood of the top origin.

Example 4.5 – Product

If M is a m -dimensional manifold and N is a n -dimensional manifold, then $M \times N$ is a $(m + n)$ -dimensional manifold. It is easy to check that the product of Hausdorff spaces is Hausdorff. If $(x, y) \in M \times N$, let $x \in U \subseteq M$ and $y \in V \subseteq N$ be Euclidean neighbourhoods, so $U \cong \mathbb{R}^m$ and $V \cong \mathbb{R}^n$. Hence, $(x, y) \in U \times V \cong \mathbb{R}^m \times \mathbb{R}^n \cong \mathbb{R}^{m+n}$.

Example 4.6 – Torus

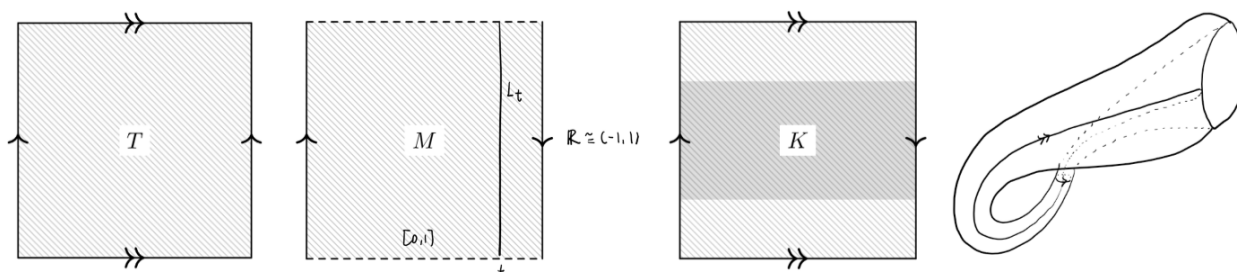
The torus $T = S^1 \times S^1$ is a 2-dimensional manifold. Using $S^1 = \mathbb{R}/\mathbb{Z}$ where $\mathbb{Z}$ represents the equivalence relation in Example 4.3, we have $T = (\mathbb{R} \times \mathbb{R})/(\mathbb{Z} \times \mathbb{Z})$. That is, the quotient of $\mathbb{R}^2$ by $x \sim y$ iff $x - y \in \mathbb{Z}^2$. Using $S^1 = [0, 1]/(0 \sim 1)$, we have $T = ([0, 1] \times [0, 1])/\approx$ where $(0, t) \approx (1, t)$ and $(s, 0) \approx (s, 1)$ for all $s, t \in [0, 1]$. In particular, T is compact since $[0, 1] \times [0, 1]$ is compact.

Example 4.7 – Möbius band

Consider the equivalence relation $\sim$ on $[0, 1] \times \mathbb{R}$, given by $(0, s) \sim (1, -s)$. Set $M = ([0, 1] \times \mathbb{R})/\sim$ to be the Möbius band. This is a 2-dimensional topological manifold. To see this, consider the subspaces $U_t = M \setminus L_t$ where $L_t = \pi(\{t\} \times \mathbb{R})$ with $0 < t < 1$. Note that $\pi^{-1}(L_t) = \{t\} \times \mathbb{R} \subset [0, 1] \times \mathbb{R}$ is closed, so $L_t \subset M$ is closed, so $U_t \subset M$ is open. Observe that $U_t = ([0, t) \times \mathbb{R} \sqcup (t, 1] \times \mathbb{R})/\sim$. Under the homeomorphism $\varphi : (t, 1] \times \mathbb{R} \rightarrow (t, 1] \times \mathbb{R}$ defined by $(x, v) \mapsto (x, -v)$, the relation $\sim$ translates to $\approx$ on $[0, t) \times \mathbb{R} \sqcup (t, 1] \times \mathbb{R}$, namely $(0, s) \approx (1, s)$. Thus, $\text{id} \sqcup \varphi$ gives a homeomorphism

$$([0, t) \times \mathbb{R} \sqcup (t, 1] \times \mathbb{R})/\sim \rightarrow ([0, t) \times \mathbb{R} \sqcup (t, 1] \times \mathbb{R})/\approx.$$

By question 10 on example sheet 1, this is homeomorphic to $(0, 1) \times \mathbb{R}$. Hence, $U_t \cong (0, 1) \times \mathbb{R} \cong \mathbb{R}^2$. As any $[(x, y)] \in M$ has $x \neq t$ for some $0 < t < 1$, $[(x, y)] \in U_t$ a Euclidean neighbourhood. Any two points in M also lie in some U_t , so Hausdorffness follows from Hausdorffness of $\mathbb{R}^2$.



Example 4.8 – Klein bottle

Combining Example 4.6 and Example 4.7, consider the relation $\approx$ on $[0, 1] \times [0, 1]$ given by $(0, t) \approx (1, 1 - t)$ and $(s, 0) \approx (s, 1)$ for all $s, t \in [0, 1]$. Let $K = ([0, 1] \times [0, 1])/\approx$ be the Klein bottle. The same considerations show that this is a 2-dimensional manifold.

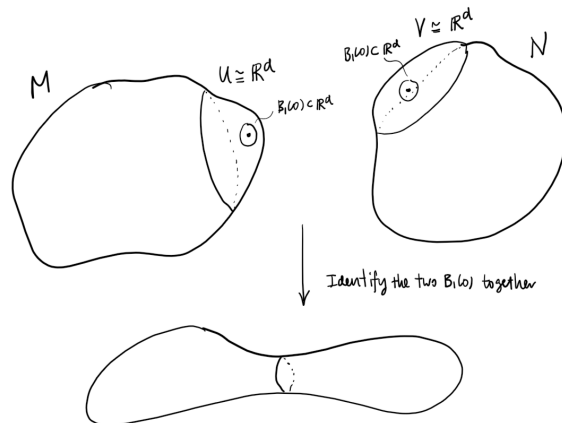
Example 4.9 – Connected-sum

Definition 4.2 – Connected sum

Let M and N be d -dimensional manifolds. Choose $m \in M$ and $n \in N$. Let $U \ni m$ be a Euclidean neighbourhood with $\varphi : U \rightarrow \mathbb{R}^d$. Let $V \ni n$ be a Euclidean neighbourhood with $\psi : V \rightarrow \mathbb{R}^d$. The connected-sum of M and N is the

$$M \# N = ((M \setminus \varphi^{-1}(B_1(0))) \sqcup (N \setminus \psi^{-1}(B_1(0)))) / \sim$$

where $\varphi^{-1}(v) \sim \psi^{-1}(v)$ for all $v \in S^{d-1} \subset \mathbb{R}^d$.



$M \# N$ is covered by three open sets. The obvious ones are $M \setminus \varphi^{-1}(\overline{B_1(0)})$ and $N \setminus \psi^{-1}(\overline{B_1(0)})$, which are open sets of topological manifolds, so are manifolds. We also have $((U \setminus \varphi^{-1}(B_1(0))) \sqcup (V \setminus \psi^{-1}(B_1(0)))) / \sim$. Using $U \setminus \varphi^{-1}(B_1(0)) \cong \mathbb{R}^d \setminus B_1(0)$ and $V \setminus \psi^{-1}(B_1(0)) \cong \mathbb{R}^d \setminus B_1(0)$, and noting that $\mathbb{R}^d \setminus B_1(0) \cong S^{d-1} \times (-\infty, 0] \cong S^{d-1} \times [0, \infty)$, we have

$$\begin{aligned} ((U \setminus \varphi^{-1}(B_1(0))) \sqcup (V \setminus \psi^{-1}(B_1(0)))) / \sim &\cong (S^{d-1} \times (-\infty, 0] \cup S^{d-1} \times [0, \infty)) / ((v, 0) \sim (v, 0)) \\ &\cong S^{d-1} \times (-\infty, \infty) \end{aligned}$$

which is a manifold. It thus follows that $M \# N$ is locally Euclidean, and we can also check that it is Hausdorff. Hence, $M \# N$ is a manifold.

Example 4.10 – Real projective space

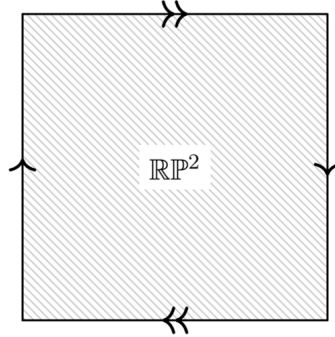
Let $\mathbb{RP}^d = \{1\text{-dimensional subspaces of } \mathbb{R}^{d+1}\}$ be a set. We give this a topology as a quotient of $\mathbb{R}^{d+1} \setminus \{0\}$ by $v \sim u$ if and only if $\langle v \rangle = \langle u \rangle$. We write $[x_0 : x_1 : \dots : x_d]$ for the equivalence class of $(x_0, x_1, \dots, x_d) \in \mathbb{R}^{d+1} \setminus \{0\}$. Thus, the sets $A_i = \{[x_0 : \dots : x_d] \mid x_i \neq 0\}$ cover $\mathbb{RP}^d$. The preimage of A_i under the quotient map is $\{(x_0, \dots, x_d) \in \mathbb{R}^{d+1} \setminus \{0\} \mid x_i \neq 0\}$, which is an open subset, so A_i is open in $\mathbb{RP}^d$. The function $\pi|_{P_i} : P_i = \{(x_0, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_d) \in \mathbb{R}^{d+1} \setminus \{0\}\} \rightarrow A_i$ is a bijection. We can check that this is a homeomorphism. Also, $P_i \cong \mathbb{R}^d$ by omitting the i -th entry. So, the A_i are Euclidean. Since every point in $\mathbb{RP}^d$ is in some A_i , the space is locally Euclidean.

To see that $\mathbb{RP}^d$ is Hausdorff, let $p, q \in \mathbb{RP}^d$. The group $\text{GL}_{d+1}(\mathbb{R})$ of linear isomorphism of $\mathbb{R}^{d+1}$ acts on $\mathbb{RP}^d$ continuously. We can find a matrix sending p and q to $\langle (1, 0, \dots) \rangle$ and $\langle (1, 1, 0, \dots) \rangle$ respectively. Let $e_1 = (1, 0, \dots)$ and $e_2 = (1, 1, 0, \dots)$, so we need to show that $\langle e_1 \rangle$ and $\langle e_2 \rangle$ can be separated. But these are points in $A_1 \cong \mathbb{R}^d$ so they can be separated by open sets. Hence, $\mathbb{RP}^d$ is a d -dimensional manifold.

Any line in $\mathbb{R}^{d+1}$ is spanned by a unit vector, well-defined up to ± 1 . Thus, we can describe $\mathbb{RP}^d = S^d / (v \sim -v)$. This shows that $\mathbb{RP}^d$ is compact since S^d is compact. Every equivalence class

can be described by an element in the northern hemisphere $D_+^d \subset S^d$, but if v is on the equator, we still have $v, -v \in D_+^d$. Hence, $\mathbb{RP}^d = D^d / (v \approx -v \text{ for } v \in \partial D^d)$ where D^d is the d -dimensional disc.

Consider the case $d = 2$. The disc is homeomorphic to the square, so we can identify points on the boundary of $[0, 1] \times [0, 1]$ as shown. $\mathbb{RP}^2$ is known as the projective plane.



Theorem 4.1

A topological manifold is connected if and only if it is path-connected.

Proof. Path-connectedness implies connectedness for all topological spaces. Let M be a connected d -dimensional manifold. The path components of M partition it into disjoint subspaces. It suffices to show that the path components are open, as then they are also all closed (since their complement is a union of open path components), so they will disconnect the space if there is more than one path component, thus there is only one and M would be path-connected.

Let $C \subseteq M$ be a path component and $c \in C$. Let $c \in U \cong \mathbb{R}^d$ be a Euclidean neighbourhood. As $\mathbb{R}^d$ is path-connected, there is a path from any $c' \in U$ to c , so $c \in U \subseteq C$ is open, hence C is open. $\square$

Theorem 4.2

If X is a connected topological manifold and $p, q \in X$, then there is a homeomorphism $\varphi : X \rightarrow X$ such that $\varphi(p) = q$. That is, the symmetry group of X (the group of all self-homeomorphisms of X) acts transitively on X . This is known as homogeneous.

Proof. We require the following lemma.

Lemma 4.1

If $p, q \in B_1(0) \subset \mathbb{R}^d$, then there is a homeomorphism φ of $\mathbb{R}^d$ which sends p to q , and is the identity outside of $B_1(0)$.

Proof. Suppose without loss of generality that $p = 0$. We can assume this since if there is $\varphi(0) = p$ and $\psi(0) = q$, then $\psi \circ \varphi^{-1}$ sends p to q . Consider first the homeomorphism

$$\begin{aligned} \rho : \mathbb{R}^d &\rightarrow B_1(0) \\ x &\mapsto \frac{x}{1+|x|} \end{aligned}$$

with inverse $y \mapsto y/(1-|y|)$. There is a homeomorphism $x \mapsto x + \rho^{-1}(q)$ of $\mathbb{R}^d$. When conjugated by ρ , it gives a homeomorphism

$$\begin{aligned} \varphi'' : B_1(0) &\rightarrow B_1(0) \\ y &\mapsto \rho(\rho^{-1}(y) + \rho^{-1}(q)) \end{aligned}$$

which sends $y = 0$ to q . Define a function

$$\begin{aligned} \varphi' : \overline{B_1(0)} &\rightarrow \overline{B_1(0)} \\ y &\mapsto \begin{cases} \varphi''(y) & \text{if } |y| < 1 \\ y & \text{if } |y| = 1. \end{cases} \end{aligned}$$

We claim that φ' is continuous. We use the sequential definition of continuity. Let (y_n) be a sequence in $B_1(0)$ converging to $y_\infty \in \partial B_1(0)$, so $|y_n| \rightarrow 1$, and $1 - |y_n|$ decreases to 0. Then,

$$\varphi'(y_n) = \varphi''(y_n) = \rho(\rho^{-1}(y_n) + \rho^{-1}(q)) = \frac{\frac{y_n}{1-|y_n|} + \frac{q}{1-|q|}}{1 + \left| \frac{y_n}{1-|y_n|} + \frac{q}{1-|q|} \right|} = \frac{y_n + \frac{1-|y_n|}{1-|q|}q}{1 - |y_n| + \left| y_n + \frac{1-|y_n|}{1-|q|}q \right|} \rightarrow y_\infty = \varphi'(y_\infty)$$

so φ' is continuous. It is a bijection. Since $\overline{B_1(0)}$ is compact and Hausdorff, φ' is in fact a homeomorphism. So,

$$\begin{aligned} \varphi : \mathbb{R}^d &\rightarrow \mathbb{R}^d \\ y &\mapsto \begin{cases} \varphi'(y) & \text{if } |y| \leq 1 \\ y & \text{if } |y| \geq 1 \end{cases} \end{aligned}$$

is continuous by the gluing lemma. The same applies to φ^{-1} , hence ψ is a homeomorphism with the required properties. $\square$

Consider the equivalence relation $x \sim y$ if and only if there exists a homeomorphism $\varphi : X \rightarrow X$ such that $\varphi(x) = y$. It partitions X into equivalence classes. As X is connected, it suffices to show that the equivalence classes are open (similar to the proof of Theorem 4.1). Let $p \in X$ and $U \ni p$ be a Euclidean neighbourhood, so we have $\tau : U \rightarrow \mathbb{R}^d$ sending p to 0. Let $C = \tau^{-1}(\overline{B_1(0)})$ so $\mathring{C} = \tau^{-1}(B_1(0))$. Lemma 4.1 says that for any $q \in \mathring{C}$, there is a homeomorphism φ_q of $\mathbb{R}^d$ sending 0 to $\tau(q)$, and such that φ_q is the identity outside $B_1(0)$. Then, $\tau^{-1} \circ \varphi_q \circ \tau$ is a homeomorphism of U sending p to q , and is the identity outside of C . Define

$$\begin{aligned} \psi_q : X &\rightarrow X \\ x &\mapsto \begin{cases} (\tau^{-1} \circ \varphi_q \circ \tau)(x) & \text{if } x \in U \\ x & \text{if } x \notin C. \end{cases} \end{aligned}$$

As X is Hausdorff, the compact subspace $C \subset U \subset X$ is closed, so $X \setminus C$ is open. Thus, ψ_q is continuous by the open set version of the gluing lemma. The same applies to ψ_q^{-1} , so ψ_q is a homeomorphism of X sending $p \in X$ to $q \in \mathring{C}$. So, the open neighbourhood $\mathring{C} \ni p$ lies in the equivalence class of p , hence the equivalence class is open. $\square$

We see that compact Hausdorff spaces are pleasant to work with. Whether a subspace is closed does not depend on the larger space around it. Closedness is an absolute property of that subspace in a compact Hausdorff space.

Example 4.11

Let $X = S^1 \sqcup \mathbb{R}$. There is no homeomorphism $\varphi : X \rightarrow X$ sending $p \in S^1$ to $q \in \mathbb{R}$. If there were, it would induce a homeomorphism $X \setminus \{p\} \rightarrow X \setminus \{q\}$. But $X \setminus \{p\}$ has two path components while $X \setminus \{q\}$ has three path components. So the conclusion of Theorem 4.2 does not hold.

Theorem 4.3

Let X be a compact d -dimensional manifold. Then, there is a $N \gg 0$ and a $Y \subset \mathbb{R}^N$ such that $X \cong Y$.

Proof. Let

$$\begin{aligned} \sigma : \mathbb{R}^d &\rightarrow [0, \infty) \\ x &\mapsto \begin{cases} 1 & \text{if } |x| \leq 1 \\ 2 - |x| & \text{if } 1 \leq |x| \leq 2 \\ 0 & \text{if } |x| \geq 2 \end{cases} \end{aligned}$$

which is continuous. For $x \in X$, there is a Euclidean neighbourhood $U_x \ni x$ and $\tau_x : U_x \rightarrow \mathbb{R}^d$. Let $V_x = \tau_x^{-1}(B_1(0)) \subset U_x$ be a smaller neighbourhood. Consider $\sigma_x = \sigma \circ \tau_x : U_x \rightarrow [0, \infty)$. Define

$$\begin{aligned} b_x : X &\rightarrow \mathbb{R}^d \\ y &\mapsto \begin{cases} \sigma_x(y)\tau_x(y) & \text{if } y \in U_x \\ 0 & \text{if } y \notin U_x \end{cases} \end{aligned}$$

which is continuous by the gluing lemma. Similarly, we can consider $\sigma_x : X \rightarrow [0, \infty)$ by letting it to be zero outside U_x . The open sets $\{V_x\}_{x \in X}$ cover X , so there is a finite subcover $\{V_{x_1}, \dots, V_{x_n}\}$. Consider

$$\begin{aligned} e : X &\rightarrow (\mathbb{R}^d \times \mathbb{R})^n =: \mathbb{R}^N \\ y &\mapsto ((b_{x_1}(y), \sigma_{x_1}(y)), \dots, (b_{x_n}(y), \sigma_{x_n}(y))) \end{aligned}$$

which is continuous by the definition of the product topology.

We show that e is injective. Suppose $e(y) = e(y')$. We have $y \in V_{x_i}$ for some i , so $\sigma_{x_i}(y) = 1$. So, $\sigma_{x_i}(y') = 1$ which means $y' \in V_{x_i}$. Since $b_{x_i}(y) = b_{x_i}(y')$, we have $\tau_{x_i}(y) = \tau_{x_i}(y')$ and so $y = y'$ as τ_{x_i} is a homeomorphism. Hence, $e : X \rightarrow \text{im}(X) \subseteq \mathbb{R}^N$ is a continuous bijection from a compact space to a Hausdorff space, so it is a homeomorphism. $\square$

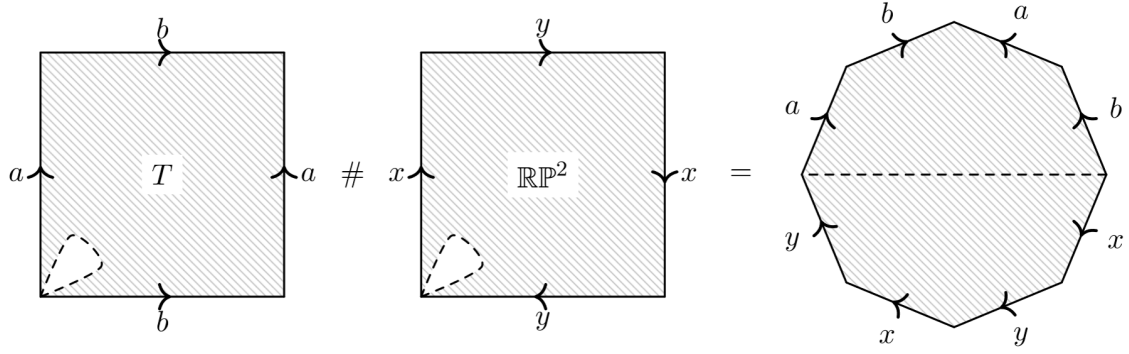
A surface is the typical name for a 2-dimensional topological manifold. We saw how to make surfaces by gluing sides of a square. More generally, for a convex polygon $P \subset \mathbb{R}^2$ having $2n$ edges, each labelled by $a, b, c, \dots$ such that every label appears exactly twice, and each marked with an arrow, we can define an equivalence relation on P by identifying points on the two edges named a in the direction indicated. Formally, if the edges called a have vertices x and y with arrow $x \rightarrow y$, and vertices u and v with arrow $u \rightarrow v$, then we can define $tx + (1 - t)y \sim tu + (1 - t)v$.

Theorem 4.4

$P / \sim$ is a surface.

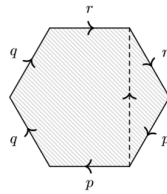
Proof. We will only provide a sketch here. To show that $P / \sim$ is locally Euclidean at $[x] \in P / \sim$, there are three cases. x is in the interior of P ; x is in the interior of an edge of P ; x is at a vertex of P . We can check for each case. $\square$

The connected-sum of surfaces can be easily described using polygons. In the example below, the bottom left corners are connected together via the connected-sum. We can replace the 8-gon by a 6-gon, by consolidating the two xy edges with two $c = xy$ edges.

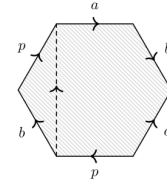
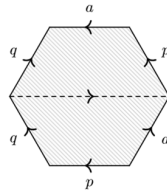


Example 4.12

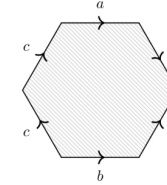
Example. In a similar way, we can obtain a 12-gon giving a connected-sum $\mathbb{RP}^2 \# \mathbb{RP}^2 \# \mathbb{RP}^2$ whose edges are all oriented clockwise and which read $xyxyststuvuv$. We may consolidate each of the contiguous edges xy , st , and uv with single edges p , q , and r to see that $\mathbb{RP}^2 \# \mathbb{RP}^2 \# \mathbb{RP}^2$ is obtained from the 6-gon



Cutting this open along the indicated dashed line a , and then gluing along r , changes this into the 6-gon



which again must give $\mathbb{RP}^2 \# \mathbb{RP}^2 \# \mathbb{RP}^2$. Cutting this open along the indicated dashed line c , and then gluing along p , changes this into the 6-gon



which again must give $\mathbb{RP}^2 \# \mathbb{RP}^2 \# \mathbb{RP}^2$. But this is the 6-gon that we explained above describes a connected-sum $T \# \mathbb{RP}^2$, so we conclude that there is a homeomorphism

$$\mathbb{RP}^2 \# \mathbb{RP}^2 \# \mathbb{RP}^2 \cong T \# \mathbb{RP}^2.$$

△

Every surface we have discussed can be triangulated in the following sense.

Definition 4.3 – Triangle, polyhedron and triangulation

A triangle in $\mathbb{R}^N$ with vertices $p, q, r \in \mathbb{R}^N$ non-collinear is

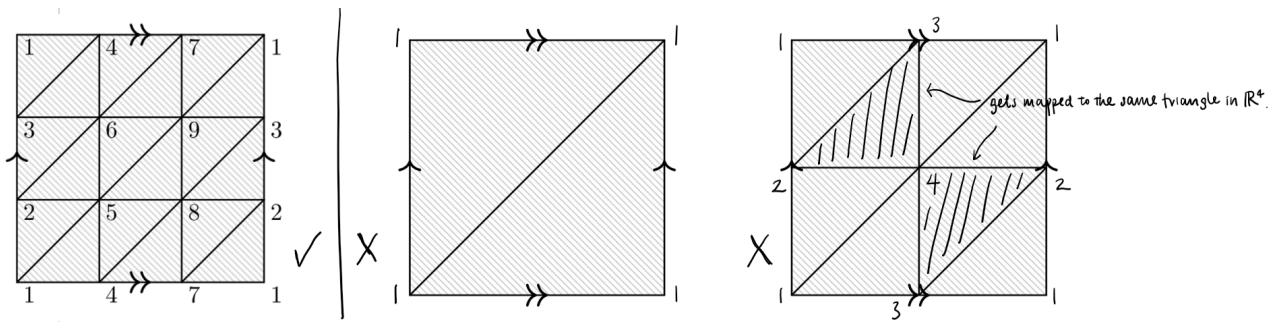
$$\{t_0p + t_1q + t_2r \in \mathbb{R}^N \mid t_0, t_1, t_2 \geq 0 \text{ and } t_0 + t_1 + t_2 = 1\}$$

which is the smallest convex set containing p, q, r . Its edges are the convex hulls of $\{p, q\}$, $\{p, r\}$, $\{q, r\}$.

A collection $K = \{\Delta_1, \Delta_2, \dots, \Delta_n\}$ of triangles in $\mathbb{R}^N$ is called a polyhedron if $\Delta_i \cap \Delta_j$ is either empty, or a common edge of Δ_i and Δ_j , or a common vertex of Δ_i and Δ_j . We set $|K| := \bigcup_{i=1}^n \Delta_i \subset \mathbb{R}^N$.

A triangulation of a surface S is the data of a polyhedron K and a homeomorphism $\varphi : |K| \rightarrow S$.

To triangulate the torus T , refer to the figure on the left. Assign the region with vertices $\{i, j, k\}$ to the triangle in $\mathbb{R}^9$ with vertices $\{e_i, e_j, e_k\}$ where the e_i 's are the basis vectors. We get a map $\varphi : T \rightarrow \mathbb{R}^9$ which is affine on each such region. We check that its image, which is a union of triangles, is a polyhedron. We can check that φ is injective (which implies that φ is a homeomorphism by compact Hausdorff) since each region is uniquely determined by its vertices.



This property is crucial. The two triangulations to the right are not valid since this property does not hold. However, this kind of naïve triangulation can always be made into a true triangulation by doing barycentric subdivision twice.

Theorem 4.5 – Radó

All compact surfaces can be triangulated.

The proof is too complicated.

Definition 4.4 – Euler characteristic

The Euler characteristic of a triangulated surface $\varphi : |K| \rightarrow S$ is the integer

$$\chi(K) = |\{\text{vertices of } K\}| - |\{\text{edges of } K\}| + |\{\text{triangles of } K\}|.$$

In Part II Algebraic Topology, we will see that if $\varphi' : |K'| \rightarrow S$ is another triangulation, then $\chi(K) = \chi(K')$, so we can say $\chi(S)$.

Example 4.13

S^2 is homeomorphic to the boundary of the tetrahedron, so $\chi(S^2) = 4 - 6 + 4 = 2$.

Example 4.14

Consider the triangulation of T before. $\chi(T) = 9 - 27 + 18 = 0$. So, the torus is not homeomorphic to the sphere.

Example 4.15

Consider $\mathbb{RP}^2$ with a triangulation similar to that of T before, but with differently labelled vertices. $\chi(\mathbb{RP}^2) = 10 - 27 + 18 = 1$. We can even calculate the Euler characteristic with fake triangulations. This is because doing a barycentric subdivision does not change χ , and we arrive at a true triangulation after two such subdivisions.

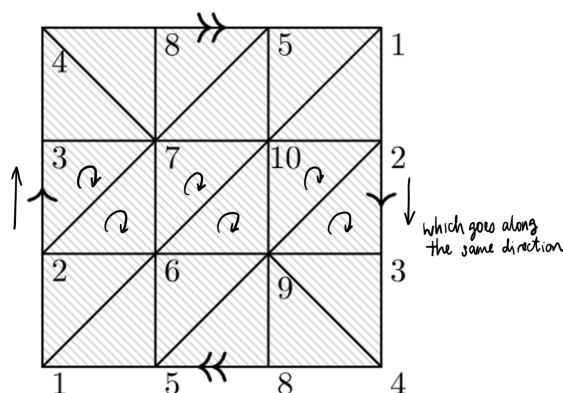
Definition 4.5 – Orientable surface

An orientation of a triangulated surface $\varphi : |K| \rightarrow S$ with polyhedron $K = \{\Delta_1, \dots, \Delta_n\}$ is a choice of cyclic ordering of the vertices of each triangle, such that if $\Delta_i \cap \Delta_j$ is an edge, then the induced ordering of the end points of this edge are opposite. If it has an orientation, then it is called orientable, otherwise it is non-orientable.

Like the Euler characteristic, orientability does not depend on the the triangulation, so it is an intrinsic property of the surface.

Example 4.16

$\mathbb{RP}^2$ has a triangulation, but it contains a Mobius band, which is non-orientable. So, $\mathbb{RP}^2$ is non-orientable. Similarly, $\mathbb{RP}^2 \# S$ for S triangulable, is non-orientable too, since we can associate the triangle $\{1, 5, 6\}$ with some part of S to form a connected sum, but it still contains the Mobius band.



Example 4.17

T is orientable. We can orient each triangle clockwise. Note that along each side, the directions are opposite its associated side, so it indeed describes an orientation.

Example 4.18

Consider $T \# T \# \dots \# T$, g times. This can be obtained by gluing edges of a $4g$ -gon with pattern $aba^{-1}b^{-1}cdc^{-1}d^{-1} \dots$. We can fake triangulate by adding a central vertices, then subdivide twice to obtain a real triangulation. Orient every triangle clockwise. As the edges $a, b, c, \dots$ appear once clockwise (a) and once anticlockwise (a^{-1}), this is a valid orientation. Hence, the surface is orientable. Similarly, S^2 is also orientable.

Theorem 4.6 – Classification of surfaces

Any compact connected surface is homeomorphic to one of

$$S^2, T, T \# T, T \# T \# T, \dots \quad \text{or} \quad \mathbb{RP}^2, \mathbb{RP}^2 \# \mathbb{RP}^2, \mathbb{RP}^2 \# \mathbb{RP}^2 \# \mathbb{RP}^2, \dots$$

and none of these are homeomorphic to each other.

Proof. An idea only. Let $\phi : |K| \rightarrow S$ be a homeomorphism. Note that each edge is only connected to two triangles. Draw the dual graph of $|K|$, that is, one vertex for each triangle, and an edge between two vertices if the two triangles share an edge. Greedily construct a maximal tree (one that does not contain any loops) of the dual graph. Glue together the abstract triangles only along the edges in the maximal tree. This gives a large polygon. The remaining edges in the dual graph tell you how to pair up the remaining edges. Thus, every connected compact surface is obtained by gluing pairs of edges of a polygon. Invent some combinatorial moves to take any polygon with edge identifications to a simpler polygon. $\square$

The first list are orientable surfaces, and have Euler characteristic $2, 0, -2, -4, \dots$ since $\chi(S_1 \# S_2) = \chi(S_1) + \chi(S_2) - 2$. The second list are non-orientable, and have Euler characteristic $1, 0, -1, -2, \dots$

Corollary 4.1

Two connected compact surfaces are homeomorphic if and only if they have the same orientability and Euler characteristic.